

MAILTRACE

Upload a saved email. See where it likely came from.

SIH26106 | AI-Powered Email Threat Detection, GeoLocation and Forensic Intelligence

Not a spam filter. Analyst lab. Localhost. No Gmail.

Two jobs. Both required.

1. Is this mail dangerous?

Spoof / phish / BEC, with reasons you can point at.

2. Where did it probably come from?

Earliest reliable hop: city + ISP. Never a person's GPS.

A TF-IDF spam model only does job 1. MailTrace is a case file.

How: parse .eml -> SPF/DKIM/DMARC stamps -> hop map -> explainable score 0-100

-> campaign graph (shared domain / IP / Reply-To) -> SHA-256 PDF evidence.

Demo corpus: 8 crafted mails (clean, spoof, lookalike, SPF fail, BEC, phish, two-campaign).

The lab. Localhost. Click a sample.

MAILTRACE

SIH26106 · analyst lab
localhost · no Gmail ingest

Drop a saved email. Get a case file.

.eml upload or a demo sample. Origin = hosting city, not a person.

Upload .eml

Choose File No file chosen

01_clean.eml

02_display_spoof.eml

03_lookalike.eml

04_spf_fail.eml

05_bec_invoice.eml

06_cred_phish.eml

07_cloud_hops.eml

08_campaign_twin.eml

Live run: display-name spoof. Score + hops + Frankfurt AWS.

07_cloud_hops.eml

08_campaign_twin.eml

SPOOF · 02_display_spoof.eml

id 5b65ebc2 · SPF fail · DKIM none · DMARC fail

CLAIMS TO BE

PEC Principal <random.account@gmail.com>

REPLY-TO

verify-now@mail-secure.net ≠ From

REASONS

1. Visible From is Gmail wearing an official title.

2. Reply-To domain does not match From.

3. Return-Path domain does not match From.

4. SPF fail. This server is not allowed to send as the claimed domain.

5. DMARC fail.

6. Young domain in offline cache: mail-secure.net

OFFLINE DOMAIN CACHE

gmail.com: 7000d · mail-secure.net: 12d (young)

Download case PDF

RECEIVED CHAIN

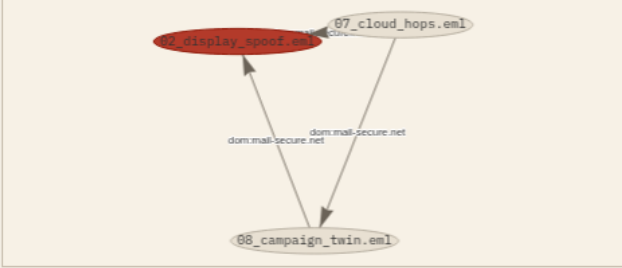
from ec2-18-184.eu-central-1.compute.amazonaws.com (ec2-18-184.eu-central-1.compute.amazonaws.com [18.184.10.20]) by mx.google.com with ESMTPS id b2; Sun, 30 Aug 2026 12:10:00 +0530

18.184.10.20 · Frankfurt · AWS

ORIGIN MAP



CAMPAIGN GRAPH



SHA-256 2fbbe59fe11629347cf6c376b5260c3a9ca0563fc238e6d395772949640039ad

Geo is hosting city / ISP of the earliest reliable hop. Not a person's GPS.

Two related mails share Reply-To + hop IP.

SPOOF · 08_campaign_twin.eml

id 9edc562f · SPF fail · DKIM none · DMARC fail



CLAIMS TO BE

Campaign Desk <followup@mail-secure.net>

REPLY-TO

camp@mail-secure.net

REASONS

1. SPF fail. This server is not allowed to send as the claimed domain.
2. DMARC fail.
3. Young domain in offline cache: mail-secure.net
4. First public hop is AWS, Frankfurt — cloud, not campus mail.

OFFLINE DOMAIN CACHE

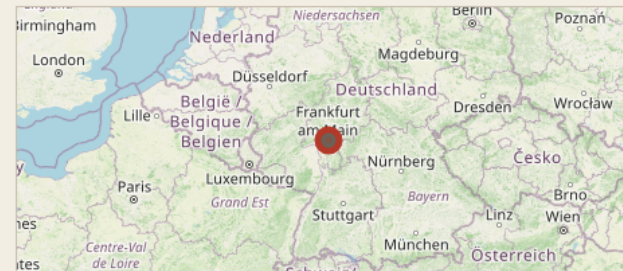
mail-secure.net: 12d (young)

[Download case PDF](#)

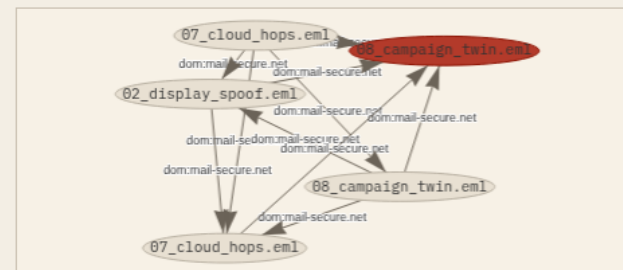
RECEIVED CHAIN

- from mx.google.com (mx.google.com [8.8.8.8]) by mailbox.pec.edu.in with ESMTPS id h8a
8.8.8.8 · unknown · public-dns
- from ec2-18-184.eu-central-1.compute.amazonaws.com (ec2-18-184.eu-central-1.compute.amazonaws.com [18.184.10.20]) by mx.google.com with ESMTPS id h8b
18.184.10.20 · Frankfurt · AWS

ORIGIN MAP



CAMPAIGN GRAPH



SHA-256 bf8848a150d1c219f60c4e9f08a2d4d469abcedb7b12aa547ca1586799e74675

Geo is hosting city / ISP of the earliest reliable hop. Not a person's GPS.

90-second demo | what we will not build

Show

1. Click 04_spf_fail.eml - auth fail.
2. Click 07 then 08 - campaign graph.
3. Download case PDF - SHA-256 of that exact file.

Runs on a laptop (localhost). No public server required for idea round.

Do not

Gmail/Outlook ingest. Sending phishing. Claiming GPS of a person.

Public blockchain of mail. Live WHOIS during demo.

Private code: github.com/nexurlabs/sih